

Table 1: Causal hyperparameter-tuning battery extended with the spectral-kNN analog family (*italic* rows). All arms share one out-of-sample panel ($n = 218,934$ thirty-minute bars, 100 walk-forward chunks with a 24,000-bar warm-up halo, pooled per bar) and one comparator: OLS on HAR + calendar with no exogenous features (the production ridge path at $\alpha = 0$, exact OLS, per-bar refit), QLIKE = 0.13415. OOS- R^2 is measured against that incumbent, so > 0 means the arm beats it. DM is a Diebold–Mariano test on per-bar QLIKE loss with HAC standard errors; negative t favours the arm. The spectral-kNN arms compose a residualizer (identity, or OLS with the per-window identifiability mask) with a W -bar view of the target path concatenated to the bucket’s non-target-derived feature row, matched by Gaussian-weighted k NN either directly (passthrough) or after a d -dimensional Laplacian embedding. Best QLIKE and best OOS- R^2 in bold. The dotted rule marks where arms stop beating the incumbent: every row above it has QLIKE below 0.13415, every row below is worse. Of 32 spectral arms, 8 beat the incumbent; the best ranks 21 of 98. The empty-exog bucket (SUBGROUPS key **baseline**) is displayed as *no-exog* (*HAR only*) so it is not confused with the incumbent row; the underlying key is unchanged in the CSV. **Provenance:** the causal-tuned arms are audit-signed; the spectral-kNN arms are NOT — they were run from an unsigned spec outside the submission journal, and cells whose unpenalized design went singular ($MZ\text{-}\beta < 0.1$) are excluded.

exogenous bucket	estimator	QLIKE	MAE	MSE	OOS- R^2	MZ- β	MZ- R^2	DM vs. incumbent (t, p)
all_features	LightGBM causal-tuned	0.12604	1.222×10^{-6}	4.080×10^{-11}	0.0591	0.892	0.7231	$-15.4, 1.1 \times 10^{-53}$
all_features	XGBoost causal-tuned	0.12641	1.231×10^{-6}	4.605×10^{-11}	-0.0621	0.840	0.7008	$-16.5, 1.7 \times 10^{-61}$
all_features	ridge causal-tuned	0.12788	1.242×10^{-6}	4.586×10^{-11}	-0.0576	0.884	0.6887	$-19.9, 2.4 \times 10^{-88}$
moments	LightGBM causal-tuned	0.12816	1.230×10^{-6}	4.245×10^{-11}	0.0209	0.882	0.7134	$-11.3, 1.9 \times 10^{-29}$
all_features	ElasticNet causal-tuned	0.12827	1.245×10^{-6}	4.260×10^{-11}	0.0175	0.905	0.7077	$-19.3, 4.1 \times 10^{-83}$
liquidity	LightGBM causal-tuned	0.12840	1.236×10^{-6}	4.086×10^{-11}	0.0577	0.892	0.7227	$-14.7, 7.1 \times 10^{-49}$
moments	XGBoost causal-tuned	0.12863	1.235×10^{-6}	4.151×10^{-11}	0.0426	0.875	0.7221	$-9.0, 1.7 \times 10^{-19}$
liquidity	XGBoost causal-tuned	0.12886	1.240×10^{-6}	3.966×10^{-11}	0.0853	0.879	0.7347	$-14.0, 2.2 \times 10^{-44}$
all_features	Lasso causal-tuned	0.12907	1.244×10^{-6}	4.232×10^{-11}	0.0240	0.915	0.7079	$-13.2, 5.9 \times 10^{-40}$
market.vw	LightGBM causal-tuned	0.12963	1.247×10^{-6}	4.378×10^{-11}	-0.0097	0.862	0.7098	$-10.8, 2.1 \times 10^{-27}$
vol.demand	LightGBM causal-tuned	0.12979	1.253×10^{-6}	4.486×10^{-11}	-0.0347	0.861	0.7023	$-10.0, 1.5 \times 10^{-23}$
implied.vol	LightGBM causal-tuned	0.13001	1.262×10^{-6}	4.483×10^{-11}	-0.0339	0.840	0.7099	$-7.8, 4.5 \times 10^{-15}$
market.vw	XGBoost causal-tuned	0.13012	1.251×10^{-6}	4.495×10^{-11}	-0.0368	0.855	0.7035	$-8.1, 7.1 \times 10^{-16}$
vol.demand	XGBoost causal-tuned	0.13026	1.266×10^{-6}	4.810×10^{-11}	-0.1094	0.845	0.6840	$-8.6, 1.1 \times 10^{-17}$
sentiment	LightGBM causal-tuned	0.13038	1.254×10^{-6}	4.611×10^{-11}	-0.0633	0.867	0.6913	$-8.1, 7.6 \times 10^{-16}$
market.ew	LightGBM causal-tuned	0.13039	1.250×10^{-6}	4.321×10^{-11}	0.0034	0.867	0.7121	$-7.0, 3.2 \times 10^{-12}$
market.ew	XGBoost causal-tuned	0.13043	1.250×10^{-6}	4.337×10^{-11}	-0.0003	0.865	0.7117	$-7.3, 3.0 \times 10^{-13}$
sentiment	XGBoost causal-tuned	0.13050	1.260×10^{-6}	4.813×10^{-11}	-0.1099	0.854	0.6810	$-8.4, 3.8 \times 10^{-17}$
implied.vol	XGBoost causal-tuned	0.13077	1.263×10^{-6}	4.636×10^{-11}	-0.0693	0.842	0.6978	$-5.7, 9.4 \times 10^{-9}$
moments	ridge causal-tuned	0.13085	1.249×10^{-6}	4.260×10^{-11}	0.0176	0.937	0.7030	$-19.2, 3.3 \times 10^{-82}$
market.vw	<i>spectral-kNN OLS-masked W=12 passthrough k=100</i>	0.13096	1.248×10^{-6}	4.220×10^{-11}	0.0267	0.928	0.7068	$-17.6, 4.4 \times 10^{-69}$
moments	ElasticNet causal-tuned	0.13114	1.253×10^{-6}	4.201×10^{-11}	0.0312	0.933	0.7075	$-17.7, 5.2 \times 10^{-70}$
no-exog (HAR only)	LightGBM causal-tuned	0.13118	1.259×10^{-6}	4.508×10^{-11}	-0.0396	0.870	0.6979	$-5.6, 2.2 \times 10^{-8}$
liquidity	ridge causal-tuned	0.13120	1.274×10^{-6}	4.345×10^{-11}	-0.0020	0.898	0.7030	$-13.4, 9.5 \times 10^{-41}$
market.vw	<i>spectral-kNN OLS W=12 passthrough k=100</i>	0.13145	1.251×10^{-6}	4.222×10^{-11}	0.0263	0.930	0.7064	$-14.6, 2.9 \times 10^{-48}$
no-exog (HAR only)	XGBoost causal-tuned	0.13163	1.262×10^{-6}	4.601×10^{-11}	-0.0611	0.855	0.6956	$-5.5, 4.1 \times 10^{-8}$
moments	Lasso causal-tuned	0.13179	1.256×10^{-6}	4.224×10^{-11}	0.0258	0.939	0.7053	$-8.9, 6.5 \times 10^{-19}$
liquidity	ElasticNet causal-tuned	0.13184	1.276×10^{-6}	4.320×10^{-11}	0.0037	0.904	0.7036	$-11.5, 2.2 \times 10^{-30}$
liquidity	Lasso causal-tuned	0.13206	1.278×10^{-6}	4.336×10^{-11}	0.0001	0.901	0.7030	$-11.2, 3.9 \times 10^{-29}$
no-exog (HAR only)	<i>spectral-kNN OLS W=24 passthrough k=100</i>	0.13218	1.267×10^{-6}	4.325×10^{-11}	0.0026	0.932	0.6988	$-12.9, 4.2 \times 10^{-38}$
no-exog (HAR only)	<i>spectral-kNN OLS W=12 passthrough k=100</i>	0.13219	1.268×10^{-6}	4.291×10^{-11}	0.0104	0.931	0.7014	$-14.4, 7.7 \times 10^{-47}$
implied.vol	ridge causal-tuned	0.13252	1.262×10^{-6}	4.240×10^{-11}	0.0222	0.922	0.7063	$-11.3, 1.8 \times 10^{-29}$
implied.vol	ElasticNet causal-tuned	0.13262	1.264×10^{-6}	4.202×10^{-11}	0.0309	0.921	0.7092	$-10.6, 2.5 \times 10^{-26}$
implied.vol	Lasso causal-tuned	0.13274	1.263×10^{-6}	4.183×10^{-11}	0.0352	0.924	0.7100	$-9.8, 9.9 \times 10^{-23}$
market.vw	ridge causal-tuned	0.13291	1.267×10^{-6}	4.248×10^{-11}	0.0202	0.928	0.7049	$-9.1, 8.5 \times 10^{-20}$
market.vw	ElasticNet causal-tuned	0.13316	1.266×10^{-6}	4.226×10^{-11}	0.0254	0.934	0.7056	$-7.3, 2.3 \times 10^{-13}$
market.ew	ridge causal-tuned	0.13319	1.264×10^{-6}	4.221×10^{-11}	0.0265	0.936	0.7058	$-7.2, 5.1 \times 10^{-13}$
market.vw	Lasso causal-tuned	0.13321	1.267×10^{-6}	4.231×10^{-11}	0.0243	0.929	0.7060	$-6.4, 1.1 \times 10^{-10}$
no-exog (HAR only)	ols.har b2 cap3125	0.13332	1.278×10^{-6}	4.349×10^{-11}	-0.0030	0.933	0.6971	$-10.6, 3.7 \times 10^{-26}$
no-exog (HAR only)	ols.har b2 cap240	0.13333	1.274×10^{-6}	4.303×10^{-11}	0.0076	0.940	0.6995	$-10.5, 7.7 \times 10^{-26}$
no-exog (HAR only)	ols.har b2 cap960	0.13334	1.276×10^{-6}	4.322×10^{-11}	0.0033	0.938	0.6985	$-10.7, 1.1 \times 10^{-26}$
vol.demand	ridge causal-tuned	0.13337	1.278×10^{-6}	4.295×10^{-11}	0.0095	0.926	0.7019	$-4.1, 4.3 \times 10^{-5}$
market.ew	ElasticNet causal-tuned	0.13344	1.266×10^{-6}	4.214×10^{-11}	0.0281	0.937	0.7062	$-5.4, 5.5 \times 10^{-8}$
sentiment	ridge causal-tuned	0.13347	1.276×10^{-6}	4.276×10^{-11}	0.0139	0.934	0.7021	$-5.7, 1.3 \times 10^{-8}$
no-exog (HAR only)	ols.har b3 cap3125	0.13349	1.279×10^{-6}	4.301×10^{-11}	0.0080	0.932	0.7006	$-10.2, 2.2 \times 10^{-24}$
no-exog (HAR only)	ols.har b3 cap960	0.13350	1.276×10^{-6}	4.272×10^{-11}	0.0147	0.938	0.7019	$-9.8, 9.1 \times 10^{-23}$
market.ew	Lasso causal-tuned	0.13353	1.268×10^{-6}	4.220×10^{-11}	0.0267	0.934	0.7060	$-4.6, 4.0 \times 10^{-6}$
no-exog (HAR only)	ols.har b3 cap240	0.13359	1.273×10^{-6}	4.251×10^{-11}	0.0195	0.943	0.7029	$-8.0, 1.1 \times 10^{-15}$
vol.demand	Lasso causal-tuned	0.13359	1.276×10^{-6}	4.261×10^{-11}	0.0174	0.931	0.7036	$-3.3, 8.1 \times 10^{-4}$
vol.demand	ElasticNet causal-tuned	0.13361	1.276×10^{-6}	4.268×10^{-11}	0.0156	0.933	0.7028	$-3.4, 6.0 \times 10^{-4}$
no-exog (HAR only)	ols.har b4 cap3125	0.13361	1.276×10^{-6}	4.324×10^{-11}	0.0027	0.938	0.6983	$-8.5, 2.4 \times 10^{-17}$
no-exog (HAR only)	<i>spectral-kNN OLS W=12 spectral d=6 k=100</i>	0.13364	1.279×10^{-6}	4.337×10^{-11}	-0.0003	0.921	0.6994	$-4.6, 4.0 \times 10^{-6}$
no-exog (HAR only)	ols.har b4 cap960	0.13368	1.276×10^{-6}	4.313×10^{-11}	0.0052	0.940	0.6988	$-7.3, 2.9 \times 10^{-13}$
sentiment	ElasticNet causal-tuned	0.13377	1.276×10^{-6}	4.247×10^{-11}	0.0205	0.941	0.7034	$-3.5, 3.9 \times 10^{-4}$
no-exog (HAR only)	ols.har b4 cap240	0.13378	1.274×10^{-6}	4.302×10^{-11}	0.0079	0.945	0.6991	$-5.3, 1.0 \times 10^{-7}$
sentiment	Lasso causal-tuned	0.13390	1.277×10^{-6}	4.244×10^{-11}	0.0212	0.939	0.7038	$-2.1, 3.6 \times 10^{-2}$
no-exog (HAR only)	<i>spectral-kNN OLS W=24 passthrough k=25</i>	0.13397	1.276×10^{-6}	4.372×10^{-11}	-0.0084	0.925	0.6964	$-0.9, 3.8 \times 10^{-1}$
no-exog (HAR only)	<i>spectral-kNN OLS W=24 spectral d=6 k=100</i>	0.13399	1.274×10^{-6}	4.374×10^{-11}	-0.0087	0.930	0.6957	$-1.6, 1.1 \times 10^{-1}$
no-exog (HAR only)	<i>spectral-kNN OLS W=12 passthrough k=25</i>	0.13405	1.275×10^{-6}	4.230×10^{-11}	0.0244	0.929	0.7059	$-0.5, 6.0 \times 10^{-1}$
no-exog (HAR only)	ols.har b5 cap3125	0.13415	1.283×10^{-6}	4.336×10^{-11}	0.0000	0.930	0.6984	$0.0, 1.0 \times 10^0$
no-exog (HAR only)	ols.har b5 cap960	0.13420	1.278×10^{-6}	4.298×10^{-11}	0.0087	0.941	0.6998	$2.1, 3.6 \times 10^{-2}$
no-exog (HAR only)	ols.har b5 cap240	0.13422	1.277×10^{-6}	4.288×10^{-11}	0.0110	0.942	0.7004	$1.9, 5.2 \times 10^{-2}$
no-exog (HAR only)	ols.har b6 cap3125	0.13433	1.280×10^{-6}	4.325×10^{-11}	0.0025	0.940	0.6980	$3.1, 1.8 \times 10^{-3}$
no-exog (HAR only)	ols.har explicit cap3125	0.13439	1.279×10^{-6}	4.280×10^{-11}	0.0129	0.942	0.7010	$3.1, 2.0 \times 10^{-3}$
no-exog (HAR only)	ols.har explicit cap960	0.13439	1.279×10^{-6}	4.280×10^{-11}	0.0129	0.942	0.7010	$3.1, 2.0 \times 10^{-3}$
no-exog (HAR only)	ols.har b6 cap960	0.13443	1.280×10^{-6}	4.311×10^{-11}	0.0058	0.942	0.6988	$4.7, 2.7 \times 10^{-6}$
no-exog (HAR only)	ols.har b6 cap240	0.13443	1.280×10^{-6}	4.311×10^{-11}	0.0058	0.942	0.6988	$4.7, 2.7 \times 10^{-6}$
no-exog (HAR only)	ridge causal-tuned	0.13445	1.282×10^{-6}	4.301×10^{-11}	0.0081	0.934	0.7004	$5.0, 4.7 \times 10^{-7}$
no-exog (HAR only)	ols.har explicit cap240	0.13447	1.279×10^{-6}	4.268×10^{-11}	0.0156	0.944	0.7016	$3.9, 9.8 \times 10^{-5}$
no-exog (HAR only)	ElasticNet causal-tuned	0.13471	1.281×10^{-6}	4.266×10^{-11}	0.0162	0.938		